

DDR4 records broken

G.Skill's Ripjaws 4 series DDR4 RAM broke the fastest DDR4 memory world record with a 4355MHz frequency. http://dgit.in/RAM_Rec

**Encrypted USB drive**

Toshiba pulled the wraps off its new USB flash drive featuring military-grade AES 256-bit encryption

and essential compounds justify humanity's mission to Europa as the closest possible source of life in our Solar System.

Another Giant Leap for Humanity

In February of 2015, the National Aeronautics and Space Administration of the United States announced that the 2016 budget would include provisions for a robotic space mission to Jupiter's moon Europa. This marked a significant step from America to move past its "mission concept" planning for a Europa mission and shift it into high gear as a "mission directive". The Europa Clipper mission, as it is currently known, joins the European Space Agency's 2012 Jupiter Icy Moon Explorer or JUICE

which include 150 kilograms of radiation shielding, are being perfected to utilize the abundant solar energy in the region.

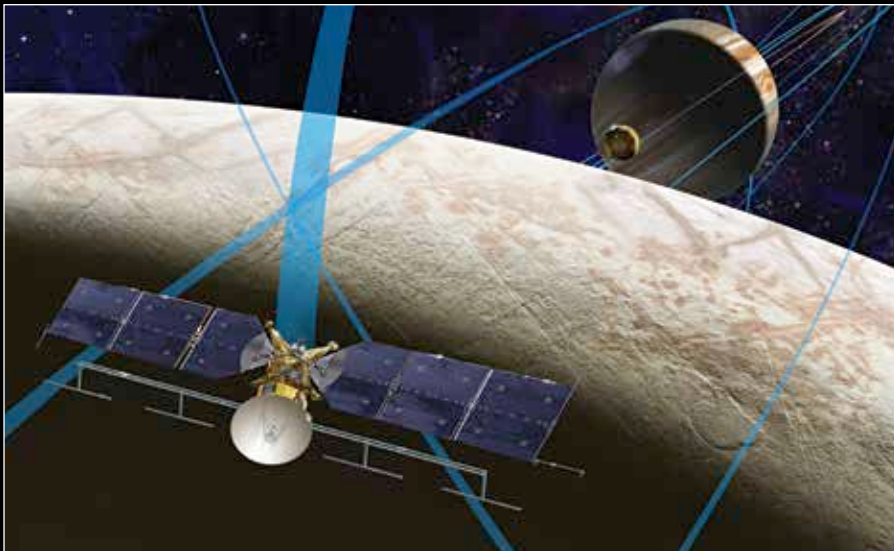
The probes within the Europa Clipper are being designed to explore the surface of the moon and determine closely its environment for habitability. The secondary mission also includes data gathering so that landing sites can be marked for future missions. The craft and its probes will actually be orbiting the planet Jupiter and will conduct flyby observations of Europa en route. The estimated number of flybys will hopefully reach 45 and will approach varying degrees of intimacy with the moon.

Although the final manifest of scientific equipment and test arrays hasn't been

Europa Clipper Explained

Travelling to Jupiter still isn't quite a walk in the park as we would hope. In the past we have sent the Voyager probes and the Galileo mission successfully within and past Jovian orbits with some measure of success. However, in the case of the Europa Clipper the mission directives are more focussed and demanding. The Europa Clipper would not only have to transverse the 778 million kilometer distance but also withstand the gigantic planets gravitational effects. To put this in context, the Earth's moon Luna is only 384,400 kilometers away, making the trip to Europa nearly 2024 times farther, and perhaps that many times more complex.

The idea of visiting Europa and other Jovian satellites has been on the books for a long time. The mission concept of the Europa mission itself took 15 years in the making. With challenges of travel distance, propulsion, fuel economy, safety and radiation risks, it hasn't been easy to plan. The heavy cost of the mission is mainly due to the fuel and radiation shielding since without the latter none of the on board electronics would be functional. And while the early missions were for shorter periods, a comprehensive mission would require the Europa Clipper to be in orbit around Jupiter for weeks, if not months. With some initial cost estimates reaching nearly USD 30 billion, it's only in the last few years that new technologies have allowed for such an ambitious and risky goal to be envisioned at affordable rates.



In 15 years, the price of the Europa Clipper concept has been reduced from USD 27 billion to USD 2 billion.

mission to visit Europa (and other moons of Jupiter). Both the missions have been on the boards for a while and have only over the last few years found financial support. The ESA JUICE mission is scheduled to launch in 2022 followed by NASA's Europa Clipper in 2025.

The Europa Clipper mission however is dedicated to the study and exploration of Europa in its entirety and has the highest likelihood of success at finding life. The overall investment by America is estimated to be over USD 2 billion and has specific goals. The yet to be built spacecraft is being designed to face the unique challenges of Jovian orbit - mainly the bombardment of radiation. The current designs for the craft,

finalised, the craft will be measuring the gravitational pull of the moon. By measuring the degree of the tugging and pulling by Europa on the Clipper, and observing the flex in the thickness of its ice shell as it orbits Jupiter, scientists will be able to confirm the presence of an ocean. If the degree of flex in response to Jupiter's gravity is high then it would mean that the ice shell isn't attached to a solid surface and is free floating while the opposite would imply that there is a solid surface below the ice. The simple success or failure of this observation alone would gain new insight into the moon that would greatly advance human endeavors towards finding life in the Solar System.

Propulsion and Power

One of the early solutions to the means of on-board power was the use of the Multi-Mission Radioisotope Thermoelectric Generators - or basically a nuclear power plant. The functionality and effectiveness of these generators had already been proven in the Mars Science Laboratory and could provide an affordable solution. In fact, of the five generators in existence, two had already been reserved for the Mars 2020 rover mission. These generators use plutonium-238 as their main fuel source which is designed to be in the form of easy to carry pellets and could be manufactured easily with some investment.